

Test 9 — Question Paper

38 items. Suggested time: 95 minutes (about 2 minutes 30 seconds per item).

A four-function calculator may be used **only** on items marked *calculator*.

The Grade 6 B.E.S.T. Mathematics Reference Sheet is on the last page.

Aligned to the FAST Grade 6 Mathematics test design: 33–42% Number Sense and Operations, 25–36% Algebraic Reasoning, 25–36% Geometric Reasoning, Data Analysis, and Probability.

1. What value of x makes the equation $x \div 2.5 = 6.8$ true? Enter your answer.

Answer: _____

Show your work

equation editor

2. Rewrite $12 + 36$ as the greatest common factor of the two numbers multiplied by the sum of two whole numbers.

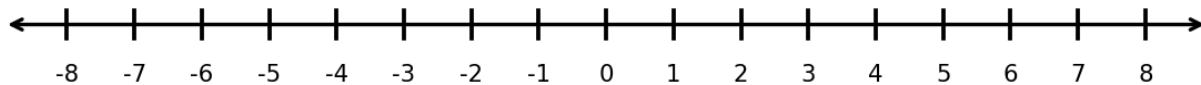
Which expression is correct?

- A. $12(1 + 4)$
- B. $1(12 + 3)$
- C. $12(4)$
- D. $12(1 + 3)$

Show your work

calculator · multiple choice

3. A football team gained 6 yards on one play and lost 5 yards on the next. Plot the two integers that represent these two plays, and plot 0.



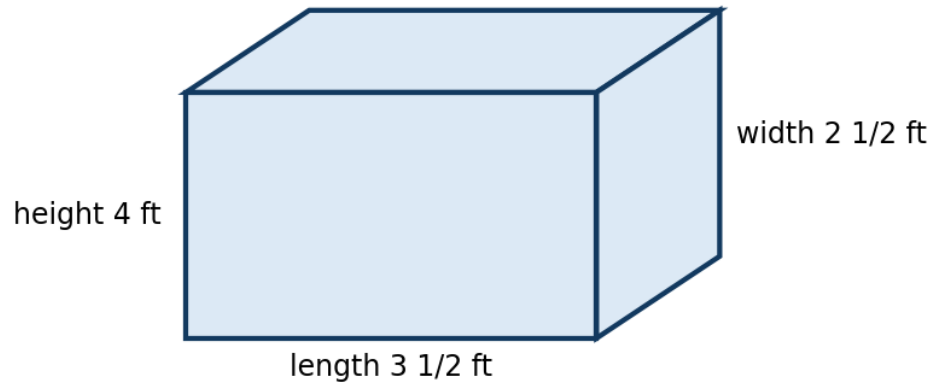
Mark your answer on the diagram above.

Show your work

calculator · graphing (grid)

4. A storage bin is a right rectangular prism with the dimensions shown.

What is its volume, in cubic feet? Enter your answer as a fraction, a mixed number or a decimal.



Answer: _____

Show your work

equation editor

5. What is the value of 2^4 ?

- A. 10
- B. 19
- C. 8
- D. 16

Show your work

calculator · multiple choice

6. Which percent is equivalent to $\frac{1}{4}$?

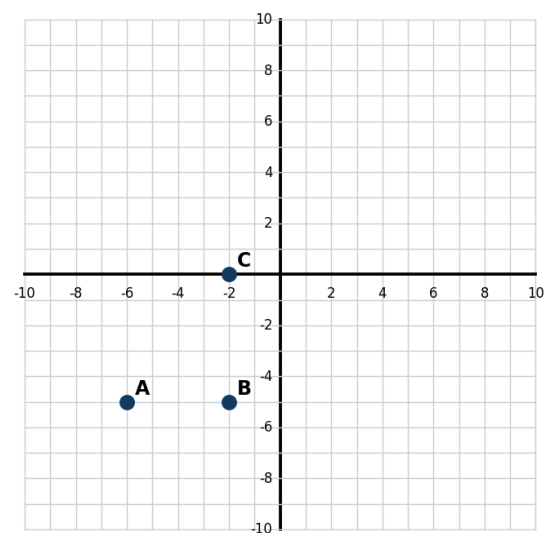
- A. 35%
- B. 2.5%
- C. 400%
- D. 25%

Show your work

calculator · multiple choice

7. Points A, B and C are three vertices of a rectangle.

Plot the fourth vertex on the coordinate plane.



Mark your answer on the diagram above.

Show your work

graphing (grid)

8. Use the drop-down menus to rewrite $84 + 60$ using the greatest common factor.

$84 + 60 = \text{[[1]]} \times (\text{[[2]]} + \text{[[3]])}$

Blank (1) choices: 2 | 7 | 12 | 24

Blank (2) choices: 5 | 7 | 8 | 84

Blank (3) choices: 5 | 6 | 7 | 60

Show your work

calculator · editing task choice

9. Use the drop-down menus to complete the equivalent expression.

$8(5x + 9) = \text{[[1]]}x + \text{[[2]]}$

Blank (1) choices: 5 | 13 | 40 | 49

Blank (2) choices: 9 | 17 | 72 | 77

Show your work

editing task choice

10. Match each description with its value.

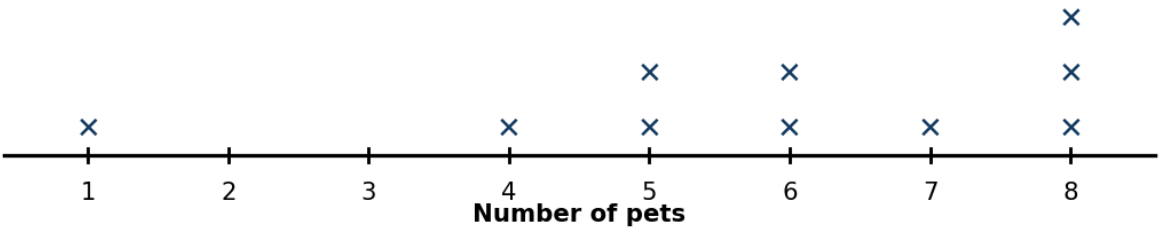
	12	24	60	72
GCF of 24 and 36	■	■	■	■
LCM of 24 and 36	■	■	■	■
GCF of 24 and 72	■	■	■	■

Show your work

calculator · matching table

11. The line plot shows number of pets reported by a group of students.

How many students reported 6 or more?



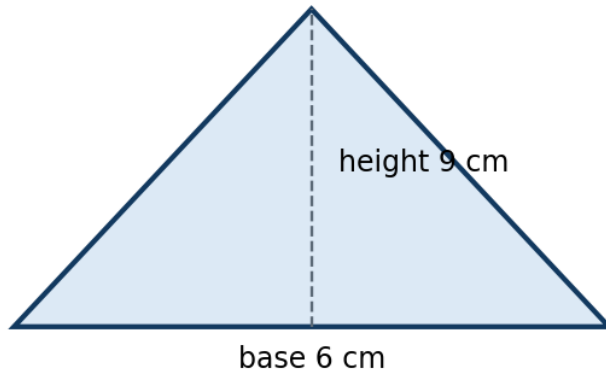
- A. 8
- B. 4
- C. 6
- D. 7

Show your work

multiple choice

12. A triangular sail has the base and height shown.

What is the area of the sail, in square centimeters? Enter your answer.



Answer: _____

Show your work

equation editor

13. There are 200 students in the sixth grade at Riviera Middle. 40% of them ride the bus.

How many students ride the bus?

- A. 160
- B. 80
- C. 5
- D. 800

Show your work

multiple choice

14. Diego has \$33.27 and spends \$9.11 on a bus pass. The rest is shared equally among 4 people.

After buying the pass, Diego has left, and each person receives .

Blank (1) choices: \$-3.17 | \$24.16 | \$42.38

Blank (2) choices: \$6.04 | \$8.32 | \$96.64

Show your work

calculator · editing task choice

15. Solve the equation $x - 15 = 37$ for x . Enter your answer.

Answer: _____

Show your work

equation editor

16. Use the drop-down menus to complete the two statements.

$34.5 \times 0.25 = [[1]]$ and $34.5 \div 0.25 = [[2]]$

Blank (1) choices: 138 | 34.25 | 34.75 | 8.625

Blank (2) choices: 138 | 34.25 | 34.75 | 8.625

Show your work

calculator · editing task choice

17. Select **all** the questions that are statistical questions.

- ☐ How tall is the tallest student in my class?
- ☐ How many text messages did each student send yesterday?
- ☐ What time does the first bell ring at my school?
- ☐ How many minutes did each sixth grader spend on the bus this morning?
- ☐ How old is my teacher?

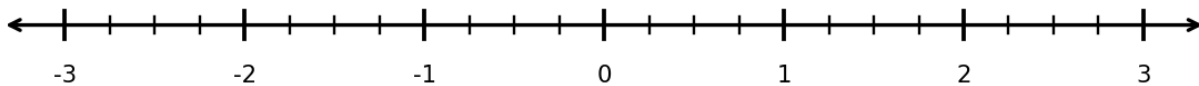
Select all that apply.

Show your work

multiselect

18. Plot these three numbers on the number line.

$-2 \frac{1}{2}$, $-1 \frac{3}{4}$, -1



Mark your answer on the diagram above.

Show your work

calculator · graphing (grid)

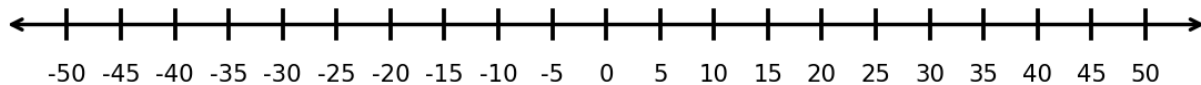
19. What is the value of $(12)(-9)$?

- A. -108
- B. 3
- C. -104
- D. 108

Show your work

calculator · multiple choice

20. Isabella and a friend swim to depths with the same absolute value. One is 35 feet below sea level. Plot both depths on the number line.



Mark your answer on the diagram above.

Show your work

calculator · graphing (grid)

21. A data set is shown.

2, 5, 8, 17, 23, 26, 27, 28, 29

Which list gives the five-number summary (minimum, lower quartile, median, upper quartile, maximum) of the data?

- A. 2, 5, 23, 28, 29
- B. 2, 23, $13\frac{1}{2}$, $55\frac{1}{2}$, 29
- C. 2, $6\frac{1}{2}$, 23, $27\frac{1}{2}$, 29
- D. 2, $13\frac{1}{2}$, 17, $55\frac{1}{2}$, 29

Show your work

multiple choice

22. A data set is shown.

11, 16, 18, 18, 22

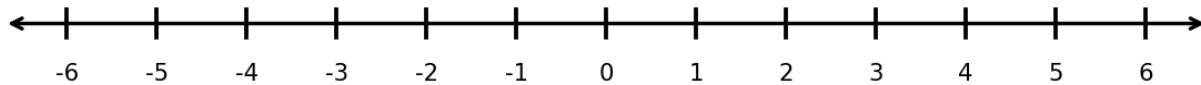
The value 32 is added to the data set. What is the mean of the new data set?

- A. 17
- B. $23 \frac{2}{5}$
- C. 32
- D. $19 \frac{1}{2}$

Show your work

multiple choice

23. Plot the two numbers on the number line that are exactly 3 units from 0.



Mark your answer on the diagram above.

Show your work

calculator · graphing (grid)

24. Which expression shows the prime factorization of 396?

- A. 2×199
- B. $2^3 \times 3^2 \times 11$
- C. $2^2 \times 3^2 \times 11$
- D. $2 \times 3^2 \times 11$

Show your work

calculator · multiple choice

25. Use the drop-down menu to complete the sentence.

The expression $x + 6$ represents 6 more than a number x .

Blank (1) choices: $6 - x$ | $x + 6$ | $x - 6$

Show your work

editing task choice

26. Solve the equation $x \div 4 = 11$ for x . Enter your answer.

Answer: _____

Show your work

equation editor

27. The table shows the points scored in five games.

What is the mean number of points? Enter your answer.

Points	#1	#2	#3	#4	#5
Points	11	11	16	17	35

Answer: _____

Show your work

equation editor

28. Match each expression with its value.

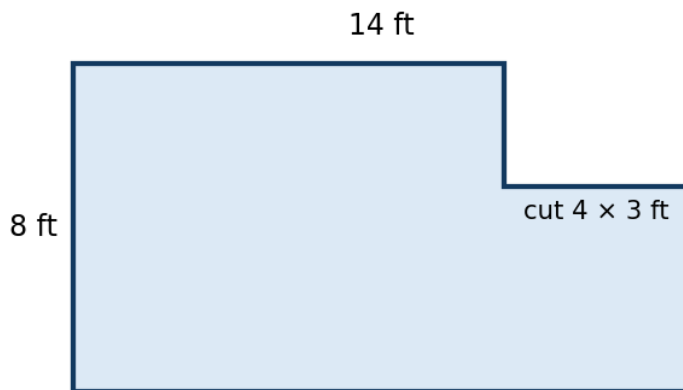
	10/27	15/8	23/18	8/15
$5/6 \times 4/9$	■	■	■	■
$5/6 \div 4/9$	■	■	■	■
$4/9 \div 5/6$	■	■	■	■

Show your work

calculator · matching table

29. The figure shows the floor of a room. All corners are right angles.

What is the area of the floor, in square feet? Enter your answer.



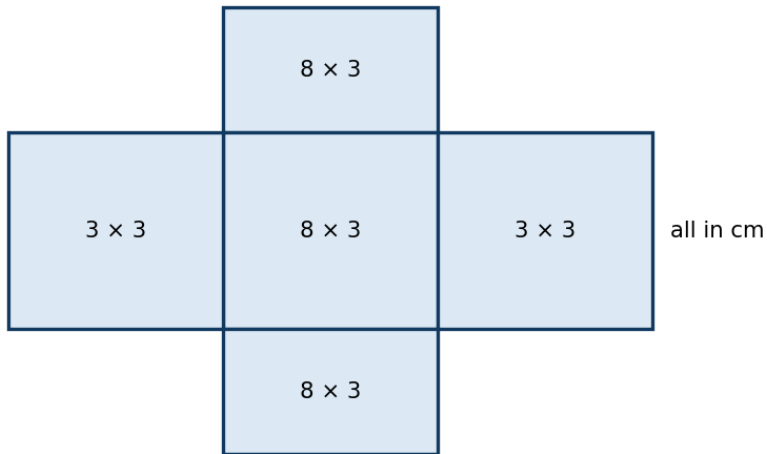
Answer: _____

Show your work

equation editor

30. A gift box is a right rectangular prism measuring 8 by 3 by 3 centimeters. Its net is shown.

How many square centimeters of paper are needed to cover the box exactly? Enter your answer.



Answer: _____

Show your work

equation editor

31. Select **all** the values that make $2x + 11 = 35$ true.

Circle every correct choice:

- ☐ 10
- ☐ 11
- ☐ 12
- ☐ 15
- ☐ 19
- ☐ 35

Show your work

selectable hot text

32. A store sells 6 pounds of guavas for \$36.00.

What is the unit price, in dollars per pound?

- A. \$1.00
- B. \$36.00
- C. \$42.00
- D. \$6.00

Show your work

multiple choice

33. The table shows equivalent ratios of red beads to blue beads.

What number belongs in the box marked ?

Red	Blue
4	7
8	14
8	?

Equivalent ratios

- A. 9
- B. 8
- C. 16
- D. 14

Show your work

multiple choice

34. What is the value of $12 + (16)$?

- A. -4
- B. -28
- C. 28
- D. 30

Show your work

calculator · multiple choice

35. Select **all** the values of x that make $x > 60$ true.

Circle every correct choice:

- ☐ 56
- ☐ 59
- ☐ 60
- ☐ 61
- ☐ 65
- ☐ 69

Show your work

selectable hot text

36. A rideshare charges a base fare of \$9.00 plus \$2.00 per mile.

What is the cost, in dollars, of a ride of $x = 3$ miles? Enter your answer.

Answer: _____

Show your work

equation editor

37. A banner is 15 feet long.

How many inches long is the banner? (1 foot = 12 inches)

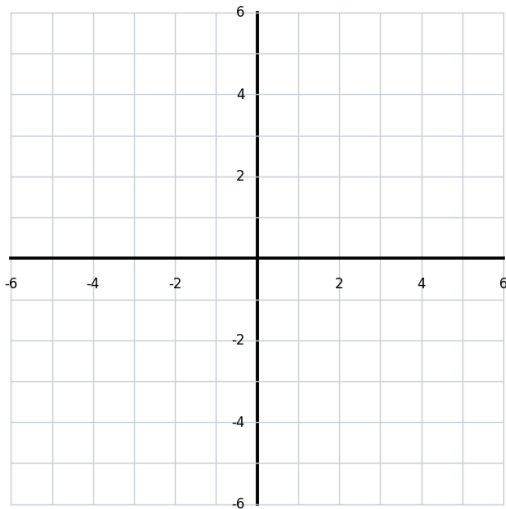
- A. 45 inches
- B. 27 inches
- C. 180 inches
- D. 1.25 inches

Show your work

multiple choice

38. Plot these three points on the coordinate plane.

$(-2, 5)$, $(2, 5)$, $(-2, -5)$



Mark your answer on the diagram above.

Show your work

graphing (grid)

Grade 6 Mathematics Reference Sheet

Customary Conversions

1 foot = 12 inches	1 cup = 8 fluid ounces
1 yard = 3 feet	1 pint = 2 cups
1 mile = 5,280 feet	1 quart = 2 pints
1 mile = 1,760 yards	1 gallon = 4 quarts
1 pound = 16 ounces	1 ton = 2,000 pounds

Metric Conversions

1 meter = 100 centimeters	1 liter = 1,000 milliliters
1 meter = 1,000 millimeters	1 gram = 1,000 milligrams
1 kilometer = 1,000 meters	1 kilogram = 1,000 grams

Time Conversions

1 minute = 60 seconds	1 week = 7 days
1 hour = 60 minutes	1 year = 365 days
1 day = 24 hours	1 year = 52 weeks

Formula

Rectangular prism	$V = lwh$ or $V = Bh$
l = length, w = width, h = height	B = area of base, V = volume